

THREE DIMENSIONAL STATISTICAL IMAGE ANALYSIS AND CONFOCAL MICROSCOPY

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ABSTRACT

We report methods for tackling a challenging 3D deconvolution problem arising in confocal microscopy. We implement Bayesian methods to produce automatic or semi-automatic segmentations showing the shape, size, orientation and spatial arrangement of objects in a sample. The methods also provide measures of uncertainty about size and shape attributes.

The 3D problem is considerably more demanding computationally than the 2D analogue considered in Al-Awadhi et al. (2004) due to the much larger data set and higher dimensional descriptors for objects in the image. In using Markov chain Monte Carlo (MCMC) simulation to draw samples from the posterior distribution, substantial computing effort can be consumed simply in reaching the main area of support of the posterior distribution. For more effective use of computation time, we use morphological techniques to help construct an initial typical image under the posterior distribution.

KEYWORDS: Bayesian statistics; Confocal microscopy; Image analysis; Markov chain Monte Carlo methods; Mathematical morphology; Object recognition; Stochastic simulation; Three-dimensional deconvolution.

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1 INTRODUCTION

In this article, we describe the fitting of Bayesian template models to 3D data in the presence of signal noise and convolution by an initially unknown point spread function. Our example is a set of confocal fluorescence microscopy data of cartilage cells. Cartilage degradation is a hallmark of some major disease groups such as osteoarthritis (Woessner and Howell, 1993) and aspects such as the types of cells, their numbers and their condition are important in determining the progress of disease. Cells are approximately ellipsoidal and can be classified into two types according to the ratios of lengths of their semi-axes.

The complexity of the statistical models for the cell configuration and for signal generation make MCMC simulation a natural choice for model fitting and inference in a Bayesian analysis. Reversible jump MCMC (Green, 1995) handles changes in dimension of the parameter vector as the number of cells increases or decreases. Such steps are made by birth, death, split or merge moves in the MCMC algorithm.

In our first attempts to implement MCMC sampling, we started from a blank image and relied on birth moves (introducing a new cell) and split moves (creating two new cells from an existing cell) to build up an accurate representation of the cell configuration. Initial progress was very slow as addition of a new cell involves a random proposal of 10 new parameter values which must match the data well enough for the step to be accepted in the algorithm while obeying the usual criteria of detailed balance. The early stage in which cells are added to the image is treated as part of the burn-in period, so the states visited do not play a role in inference about the posterior distribution. Since the major objective in this transitional phase is to locate the main area of support of the posterior distribution, the Markov chain must move towards regions of high probability but this does not require a property as precise as that of detailed balance. Once the region of support is found, one can run a proper MCMC sampler maintaining detailed balance, and allow a burn-in period for *this* Markov chain to reach equilibrium. We believe our problem is representative of other applications of MCMC simulation where the natural choice of Markov chain struggles to find the target distribution's major area of support, but mixes reasonably well once this region is found.

We describe important features of confocal microscopy and our particular data set in Section 2. We show that a simple segmentation using morphological techniques helps describe the sample of cells but note the results also demonstrate limitations of this approach. In Section 3 we present a statistical model for confocal microscope data and define a high-level image prior for use in a Bayesian analysis. We present MCMC algorithms for sampling the posterior distribution in Section 4, including a morphology-based method for creating a good initial configuration and a variation on this involving user interaction. In Section 5 we extend the model to include two distinct cell types and use this model to classify individual cells.

2 THE DATA

2.1 CONFOCAL MICROSCOPY

Confocal fluorescence microscopy is ideal for analysing three dimensional specimens since its non-destructive, non-invasive nature allows imaging of a sample at a sequence of depths without physical sectioning. Despite improvements in the technology, the images suffer from particular forms of degradation (Wilson, 1990; Pawley, 1996). The signal-to-noise ratio can be low and images appear blurred as a result of scattering of incoming and outgoing light. The signal is subject to convolution and attenuation to an extent which varies with depth. Shaw and Rawlins (1991) assess the point spread function for different types of microscope and conclude that, in most planes, it can be described by a central disc surrounded by rings of subsidiary maxima.

We concentrate in this paper on the experimental data displayed in Figure 1 which comprises a stack of two-dimensional images of an area of cartilage growth. The 2D sections at depths 3, 7 and 12 shown at a higher resolution in Figure 2 will provide a point of comparison for later reconstructions of the configuration of cells in the sample.

2.2 SEGMENTATION BY MORPHOLOGICAL TECHNIQUES

A direct approach to image segmentation for 2D or 3D data is provided by methods of mathematical morphology (Matheron, 1975; Serra, 1982). Figure 3 shows output from

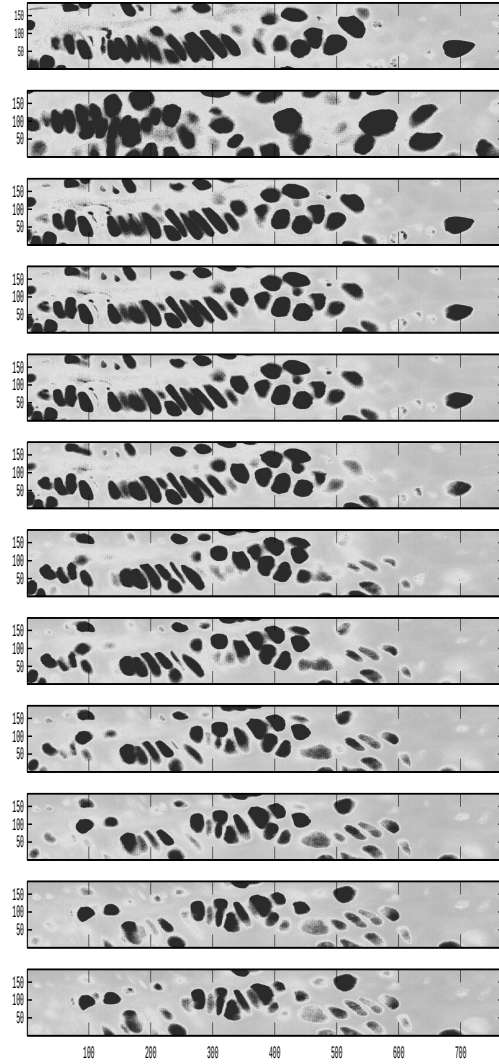


FIGURE 1: The 3D data represented as consecutive 2D optical sections.

an application of this approach. The data displayed in Figure 1 were thresholded at the median value in each optical section (as a result of attenuation, the median is lower in deeper sections), and the resulting 3D binary image was processed using an “opening” operator with a cube of $3 \times 3 \times 3$ voxels as the structuring element (see, for example, Glasbey and Horgan, 1995, Ch. 5) to produce sets of connected voxels.

The morphological approach produces a segmentation containing a number of objects, each made up of connected voxels but it does not provide an estimate of the accuracy of this segmentation. Although the output suggests cells are roughly ellipsoidal in shape, this information is not itself used in the segmentation. We shall

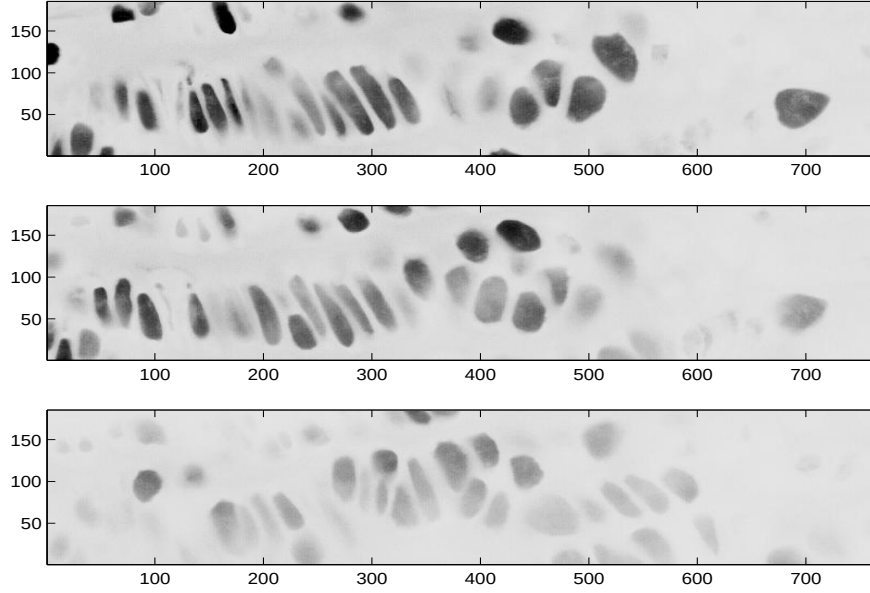


FIGURE 2: Sections of the 3D data at depths 3, 7 and 12.

present a Bayesian analysis in which the shape of objects is built into the prior model and, with a suitable model for data generation in the confocal microscope, the analysis yields an estimate of the 3D cell configuration plus a measure of confidence in this estimate. We shall, however, return to this morphological analysis as a first step in the construction of a good starting point for the MCMC simulation used to carry out the Bayesian analysis.

3 A BAYESIAN MODEL FOR CONFOCAL MICROSCOPY

3.1 THE LIKELIHOOD FUNCTION

The microscope measures low numbers of fluorescence photon counts emitted at or near the focal point, subject to scatter and attenuation effects. The counts are scaled by the operator to improve contrast, with a zero count mapping to a positive value known as the “black-level” and the highest reading set at or below a maximum permitted level. Let i, j and k index the voxels, where k represents depth in the specimen and i and j index the grid at a particular depth, and denote the signal from voxel ijk by Y_{ijk} .

We assume the background medium has typical fluorescence level τ_0 . If voxel ijk lies in a (cartilage) cell it has an increase in fluorescence level over τ_0 of λ_{ijk}

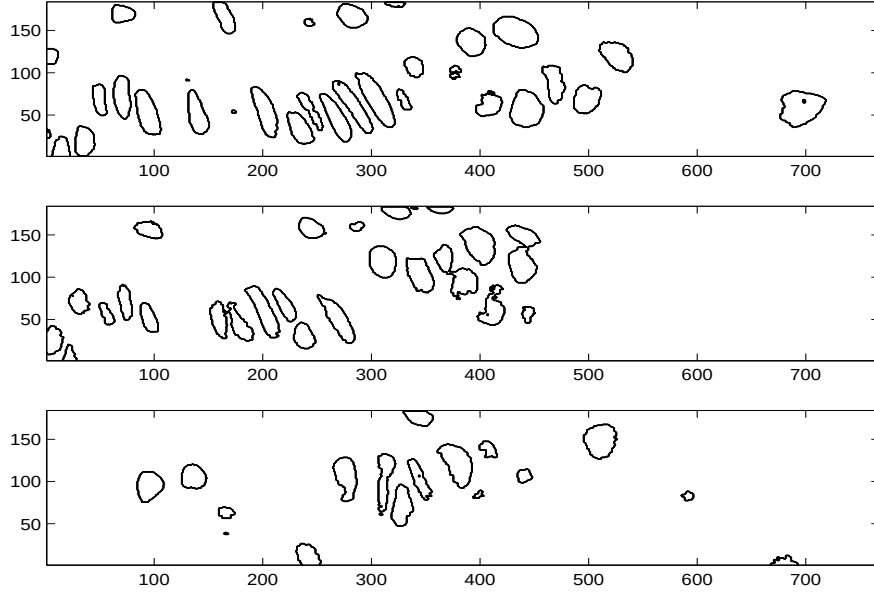


FIGURE 3: 2D sections at depths $z = 3, 7$ and 12 after applying morphological operators.

and this increase is constant for voxels lying in the same cell. Define $\lambda_{ijk} = 0$ if voxel ijk is in the background, then the fluorescence level for a general voxel ijk is $\tau_0 + \lambda_{ijk}$. Scatter causes blurring by a three-dimensional discretised Gaussian point spread function with neighbourhood Λ_{ijk} for voxel ijk and weights $\{w_{i'j'k'}\}$ summing to 1 over $i'j'k' \in \Lambda_{ijk}$. We assume a geometric form of attenuation with parameter Υ , reducing the mean count at depth k by a factor Υ^k . The user adjustment is taken to be addition of a constant d . The mean for record Y_{ijk} associated with voxel ijk is thus

$$\mu_{ijk} = \Upsilon^{k-1} \left\{ \sum_{i'j'k' \in \Lambda_{ijk}} w_{i'j'k'} (\tau_0 + \lambda_{i'j'k'}) \right\} + d.$$

We take a Gaussian approximation to the Poisson distribution and add instrument noise, modelled as additive white noise with variance ϕ_0^2 . Thus, Y_{ijk} is assumed to be a Gaussian random variable with mean μ_{ijk} and variance

$$\sigma_{ijk}^2 = \phi_0^2 + (\mu_{ijk} - d).$$

A plot of mean signal values from cell interiors against depth supports the above model for attenuation effects and indicates the value 0.945 for the attenuation factor Υ .

Examining data from the background area and central regions of cells in the light of this value of Υ led us to assign values $d = 20$, $\tau_0 = 22$ and $\phi_0^2 = 35.5$. We studied the change in signal on crossing a cell boundary in order to select an appropriate convolution function. Since the change from cell to background intensity took place over typically two to three voxels, we define Λ_{ijk} , the 3D vicinity of voxel ijk , to comprise voxels with indices within plus or minus one of i , j and k and we took the weights w_{ijk} to be from a 3 dimensional Gaussian distribution with independent components and standard deviations 0.26.

Later, we shall use X to denote a high-level representation of a configuration of cells with implied values for the μ_{ijk} and σ_{ijk}^2 defined above. We assume the records Y_{ijk} are conditionally independent given $X = x$, and the likelihood function is

$$L(y|x) = \prod_i \prod_j \prod_k \frac{1}{\sqrt{2\pi\sigma_{ijk}^2}} \exp\left\{-\frac{(y_{ijk} - \mu_{ijk})^2}{2\sigma_{ijk}^2}\right\}. \quad (1)$$

3.2 PRIOR MODEL FOR THE OBJECT CONFIGURATION

Since our objective is inference about the number of cells as well as their characteristics, we require the prior model to handle both the dimensionality of the problem and the geometric features of the list of objects: their shapes, locations and orientations. Baddeley and Van Lieshout (1993) propose using marked point process models for a random number of geometric objects. The “point” is the location of an object while the associated “mark” determines its characteristics. In our example, a realisation of the model is written as an ordered list, $X = (X_1, \dots, X_N)$, where each X_i carries information about the location, size, orientation and fluorescence of one particular cell. Note that the number of cells, N , is itself a random variable. Interactions between cells can be specified and we shall use the so-called “hard-core” model to prohibit overlap of any of the cells

The cells in our example have no completely regular shape but they are well approximated as ellipsoids. The ellipsoid centre $l = (l_x, l_y, l_z)$ is the “point” of the point process. Semi-axes a , b and c represent the width, length and height of the ellipsoid and the orientation of the ellipsoid in 3D is described through the Euler angles $\theta = (\theta_1, \theta_2, \theta_3)$ (Spiegel, 1967). These latter six parameters along with the

cell's intensity level, λ , form the “mark” of the marked point process. The value of λ_{ijk} for voxel ijk appearing in the likelihood (1) through the definitions of μ_{ijk} and σ_{ijk}^2 is the intensity of the cell containing voxel ijk or zero if the voxel lies in the background.

In specifying the prior distribution we first define a “reference model” which has the simplifying feature of no interaction between cells. The prior model and the posterior given the observed data are then written as densities with respect to the measure, Γ , of the reference model. In the reference model, cell locations l follow a homogeneous Poisson point process with unit intensity (see Diggle, 2003) in the volume under study, the marks of different cells are independent and for a particular cell a, b, c, θ and λ are independent: the semi-axes follow independent normal prior distributions $a \sim N(\mu_a, \gamma_a^2)$, $b \sim N(\mu_b, \gamma_b^2)$ and $c \sim N(\mu_c, \gamma_c^2)$, each restricted to the range $(\min_{axis}, \max_{axis})$, the Euler angles are independent with each of θ_1, θ_2 and $\theta_3 \sim U(0, \pi)$, and the fluorescence intensity $\lambda \sim U(\min_\lambda, \max_\lambda)$. Exploratory investigations led to the choice of hyperparameters $\mu_a = 25$, $\mu_b = 12.5$, $\mu_c = 3$, $\gamma_a = \gamma_b = 5.5$, $\gamma_c = 4$, $\min_{axis} = 3$, $\max_{axis} = 60$, $\min_\lambda = 40$ and $\max_\lambda = 213$.

In this reference model, the number of cells, N , has a Poisson distribution with mean $\mathcal{V} = 786 \times 184 \times 12$, the volume of the 3D image. Given $N = n$, the n cells are independent with locations distributed uniformly in this volume and other properties following the distributions described above. Let $x_i = (l_{xi}, l_{yi}, l_{zi}, a_i, b_i, c_i, \theta_{1i}, \theta_{2i}, \theta_{3i}, \lambda_i)$ be the vector of location, semi-axis lengths, Euler angles, and intensity of cell i and let $f_X(x_i)$ denote the density of x_i in $\mathbb{R}^{10}$ under the distributions specified above. The reference measure Γ is a collection of sub-measures Γ^n , $n = 0, 1, \dots$, where Γ^n concerns images with n cells defined, therefore, by $10n$ parameters. The sub-density of Γ^n with respect to Lebesgue measure on $\mathbb{R}^{10n}$ is

$$p_\Gamma^n(x) = \frac{e^{-\mathcal{V}} \mathcal{V}^n}{n!} \prod_{i=1}^n f_X(x_i).$$

Permitting cells to overlap in the reference model leads to simple expressions for the sub-densities $p_\Gamma^n(x)$. We impose the restriction to non-overlapping cells by defining the prior model to have density

$$p_X(x, n) = k \beta^n I[\text{No overlap of cells in } x] \quad (2)$$

with respect to Γ . Here β is a positive parameter reflecting the cell density, the indicator variable imposes the condition that cells do not overlap, and k is the overall normalizing constant. It follows that the prior model π_X can be written as a sum of sub-measures $\pi_X^n(x)$ with sub-densities

$$\begin{aligned} p_X^n(x) &= p_X(x, n) p_\Gamma^n(x) \\ &= k \beta^n \frac{e^{-\mathcal{V}} \mathcal{V}^n}{n!} \prod_{i=1}^n f_X(x_i) I[\text{No overlap of cells in } x] \end{aligned} \quad (3)$$

in $\mathfrak{R}^{10n}$. We set $\beta = 50/\mathcal{V}$ so the expected number of cells before applying the “no overlap” condition is 50.

3.3 THE POSTERIOR DISTRIBUTION OF THE OBJECT CONFIGURATION

The density $p_{X|Y}(x, n|y)$ of the posterior image distribution with respect to Γ is proportional to the product of the prior density and the likelihood of the observed data given x . It follows that the posterior distribution $\pi_{X|Y}$ is the sum of sub-measures $\pi_{X|Y}^n(x|y)$ with sub-densities

$$p_{X|Y}^n(x|y) \propto p_X(x, n) p_\Gamma^n(x) L(y|x), \quad n = 0, 1, \dots, \quad (4)$$

in $\mathfrak{R}^{10n}$ where $L(y|x)$ is defined by (1). Note that the normalising constant k in $p_{X|Y}^n(x|y)$ does not depend on n and hence ratios of sub-densities for different values of n can be calculated from these formulae.

4 MARKOV CHAIN MONTE CARLO SIMULATION

4.1 AN MCMC SAMPLER FOR THE POSTERIOR IMAGE DISTRIBUTION

We use Markov chain Monte Carlo methods to simulate from the posterior distribution defined by (4) and, hence, draw inferences about the cell configuration under observation. With a large number of such samples, point and interval estimates for parameters can be constructed as well as other summaries of the posterior distribution. For a general background to MCMC methods we refer readers to Besag et al. (1995) and Gilks et al. (1996).

In our MCMC sampler we include both local moves to update properties of cells and higher level moves which add cells to the current configuration or remove cells from it. We have used Metropolis-Hastings steps (Hastings, 1970) for all these move types, proposing a move and then accepting the proposal with a probability depending on the ratio of posterior densities for the existing and proposed states.

We define eight move types in total. The four types of local update select a cell at random and modify its location, size, orientation or intensity. The four types of high level move are: addition (birth) of a new cell, deletion (death) of an existing cell, merging of two adjacent cells, or splitting a cell into two. For some states, fewer than eight types of move are available, for example, if there are no cells at all in the current image, only a birth move is possible. The move type is selected from those available with equal probability for each, and we denote the probability of selecting move type ν when in state x by $j_\nu(x)$.

The “local” updates are relatively straightforward as the dimension of the state remains unchanged. In an update of location, each co-ordinate of the cell centre is perturbed independently by adding a $N(0, 4)$ variate, subject to the constraint that the location remains inside the volume under study. In updating size, an independent $U(-2, 2)$ variate is added to each semi-axis length with the constraint that lengths remain in the range $(3, 60)$. Orientation is modified by adding independent $U(-\pi/10, \pi/10)$ variates to each Euler angle then, if necessary, adding or subtracting π to bring the angle into the range $(0, \pi)$. The proposal for a new value of intensity is created by adding a $U(-10, 10)$ variate, subject to the constraint that the new value is in the range $(40, 213)$.

Let $q(x, x')$ denote the proposal density for one of these local moves defined on the appropriate subspace of $\mathbb{R}^{10n}$. The support of the reverse move from x' to x lies in the same subspace; also, the same set of move options is available at both x and x' so $j_\nu(x) = j_\nu(x')$. It follows that the Metropolis-Hastings formula for the acceptance probability of the proposal to move to x' is

$$\alpha(x, x') = \min\left\{1, \frac{p_{X|Y}^n(x'|y) q(x', x)}{p_{X|Y}^n(x|y) q(x, x')}\right\}$$

and the formulae in (1), (3) and (4) can be used to compute $p_{X|Y}^n(x'|y)$ and $p_{X|Y}^n(x|y)$.

The “high level” updates involve a change in the number of cells and, therefore, a change in the dimension of x . We have employed the “reversible jump” method of Green (1995) to do this in a manner that preserves detailed balance with respect to $\pi_{X|Y}$. We make use of the ordering in the list $X = (X_1, \dots, X_n)$ in specifying the cells involved in these updates: a birth adds a new cell X_{n+1} at the end of the list, a death removes the final cell X_n , a split replaces X_n by a new X_n and X_{n+1} , a merge replaces X_{n-1} and X_n by a new X_{n-1} . Randomly permuting elements in the list before each update ensures equal treatment of all cells. An equivalent definition can be framed in terms of a random choice of cells for death, split or merge with appropriate rules for the placement of newly created cells in the list. Alternatively, prior and posterior distributions may be defined on the space of *sets* of cells (unordered lists); see Al-Awadhi et al. (2004) for further details and remarks on the inter-relation of these approaches.

Let x be a cell configuration containing n cells. Suppose ν denotes a birth or split move which will increase the number of cells by one to $n + 1$, and let ν' be the reverse move, i.e., a death if ν is a birth or a merge if ν is a split. The move ν is implemented by generating a random variate u from a density $q_\nu(u)$ on $\mathfrak{R}^{10}$. The proposal for the new configuration with $n + 1$ cells is $x' = g_\nu(x, u)$ where the function g_ν is defined so that, for given u , a unique state x leads to each possible $x' = g_\nu(x, u)$. In the reverse moves of type ν' , indicating a death or merge update, the number of cells is reduced. With a current state x' containing $n + 1$ cells, the proposal x of an n cell configuration is specified deterministically: for a death move X_{n+1} is simply deleted, whereas for a merge move cells X_n and X_{n+1} are replaced by a new X_n , as explained below. In a birth or split move, the proposal for a transition from x to $x' = g_\nu(x, u)$ is accepted with probability

$$\alpha_\nu(x, x') = \min\left\{1, \frac{j_\nu(x') p_{X|Y}^{n+1}(x'|y)}{j_\nu(x) p_{X|Y}^n(x|y) q_\nu(u)} \left| \frac{dx'}{d(u, x)} \right| \right\}, \quad (5)$$

where $|dx'/d(u, x)|$ denotes the Jacobian of the transformation $x' = g(x, u)$. The proposal for a transition from x' to x in a death or merge move is accepted with probability

$$\alpha_{\nu'}(x', x) = \min\left\{1, \frac{j_\nu(x) p_{X|Y}^n(x|y) q_\nu(u)}{j_\nu(x') p_{X|Y}^{n+1}(x'|y)} \left| \frac{dx'}{d(u, x)} \right|^{-1} \right\},$$

where u is the unique variate satisfying $x' = g_\nu(x, u)$.

It remains to give details of the density $q_\nu(u)$ and function $g_\nu(x, u)$ when ν is a birth or split move. In the case of a birth, an additional cell X_{n+1} is added with location, semi-axis lengths, Euler angles and intensity sampled from the distribution $f_X(x_{n+1})$ arising in the reference model part of the prior model. This corresponds to taking $q_\nu(u) = f_X(u)$, $x_{n+1} = u$ and $|dx'/d(u, x)| = 1$ in (5). Note that if the new cell overlaps an existing cell $p_X(x', n+1) = 0$, hence $p_{X|Y}^{n+1}(x'|y) = 0$ and $\alpha_\nu(x, x') = 0$ so the proposal is automatically rejected.

In describing split and merge moves, it is helpful to define the merge move first. Suppose a merge move is to be proposed for cells X_A and X_B with locations l_A and l_B , semi-axis lengths a_A, b_A, c_A and a_B, b_B, c_B , Euler angles θ_A and θ_B , and intensities λ_A and λ_B . A merge is only permitted when cells are close together and share similar features: specifically, we require the distance between l_A and l_B to be at most ϵ_d , the Euler angles to satisfy

$$|\theta_{j,A} - \theta_{j,B}| \leq \epsilon_\theta \quad \text{or} \quad |\theta_{j,A} - \theta_{j,B}| \geq \pi - \epsilon_\theta, \quad \text{for } j=1, 2 \text{ and } 3,$$

and $|\lambda_A - \lambda_B| \leq \epsilon_\lambda$. We have used the values $\epsilon_d = 2.5\mu_a = 62.5$, $\epsilon_\theta = \pi/40$ and $\epsilon_\lambda = (\max_\lambda - \min_\lambda)/5 = 34.6$. Supposing these conditions are satisfied, we represent the vector $P = l_A - l_B$ between the two centres in spherical polar coordinates (d, Δ_1, Δ_2) , so $d = \|P\|$, Δ_1 is the angle between the x -axis and the projection of P in the xy -plane, and Δ_2 is the angle between the z -axis and P . The proposal for a merged cell has location $l_C = (l_A + l_B)/2$.

The alignment of cells in Figure 2 indicates that typical cases for merging will involve pairs of cells with a long semi-axis close to the direction of the y -axis, and second longest semi-axis close to the x -axis. In such cases, the vector P has $\Delta_1 \approx 0$ and $\Delta_2 \approx \pi/2$ and we have chosen our proposal for merge moves to reflect this. We define

$$\begin{aligned} a_C &= \frac{1}{2} \{1 + |\sin(\Delta_1)|\} |\sin(\Delta_2)| (a_A + a_B), \\ b_C &= \frac{1}{2} \{1 + |\cos(\Delta_1)|\} |\sin(\Delta_2)| (b_A + b_B), \\ c_C &= \frac{1}{2} \{1 + |\cos(\Delta_2)|\} (c_A + c_B). \end{aligned} \tag{6}$$

When $\Delta_1 \approx 0$ and $\Delta_2 \approx \pi/2$, this gives $a_C \approx (a_A + a_B)/2$, $b_C \approx (b_A + b_B)$ and $c_C \approx (c_A + c_B)/2$. As Δ_1 moves away from zero, a_C increases and b_C decreases. If Δ_2 moves away from $\pi/2$, the value of c_C increases, which is appropriate in cases where the two merging cells are at different depths in the z -plane.

The Euler angles for the merged cell are defined as

$$\theta_{j,C} = \begin{cases} \frac{1}{2} (\theta_{j,A} + \theta_{j,B}) & \text{if } |\theta_{j,A} - \theta_{j,B}| \leq \epsilon_\theta, \\ \{\frac{1}{2} (\pi + \theta_{j,A} + \theta_{j,B})\} \bmod \pi & \text{if } |\theta_{j,A} - \theta_{j,B}| > \pi - \epsilon_\theta, \end{cases}$$

for $j = 1, 2$ and 3 . The merged cell's intensity is set as $\lambda_C = (\lambda_A + \lambda_B)/2$.

We define the split move to produce two new cells satisfying the requirements to permit a merge move and such that application of the merge move to the two new cells returns the original cell. Starting from a single cell, labelled C , we generate a random point h from a uniform distribution within a sphere of radius $\epsilon_d/2$ and set $l_A = l_C + h$ and $l_B = l_C - h$. We generate three independent $N(0, 3)$ random variates u_a , u_b and u_c and define semi-axis lengths to satisfy simultaneously equation (6) and $a_A = c_A + u_a$, $b_A = c_A - u_a$, $a_B = c_B + u_b$, $b_B = c_B - u_b$, $a_C = c_C + u_c$, $b_C = c_C - u_c$. For the Euler angles, we generate three independent $U(-\epsilon_\theta/2, \epsilon_\theta/2)$ random variates, v_1 , v_2 and v_3 , and set

$$\theta_{j,A} = \theta_{j,C} + v_j, \quad \text{and} \quad \theta_{j,B} = \theta_{j,C} - v_j \quad \text{for } j=1, 2 \text{ and } 3,$$

adding or subtracting π as necessary to bring values into the range $(0, \pi)$. Finally, we generate a $U(-\epsilon_\lambda/2, \epsilon_\lambda/2)$ variate w and set $\lambda_A = \lambda_C + w$ and $\lambda_B = \lambda_C - w$.

The variables h , u_a , u_b , u_c , v_1 , v_2 , v_3 and w form the 10-dimensional variate u used to create a split move and the distribution $q_\nu(u)$ and function $g_\nu(u, x)$ are implied by the above definitions. Proposals that contain overlapping cells or do not meet the restrictions on ranges for a cell's location, semi-axis lengths and intensity are automatically rejected as they have probability zero in the posterior distribution. Some such proposals can be avoided by limiting the ranges of h , u_a , u_b , u_c and w , with appropriate corrections to the density $q_\nu(u)$.

Our definitions of merge and split moves have been chosen to generate proposals with a good chance of acceptance. It is permissible to take advantage of the nature of the example under consideration: other choices may be preferred in

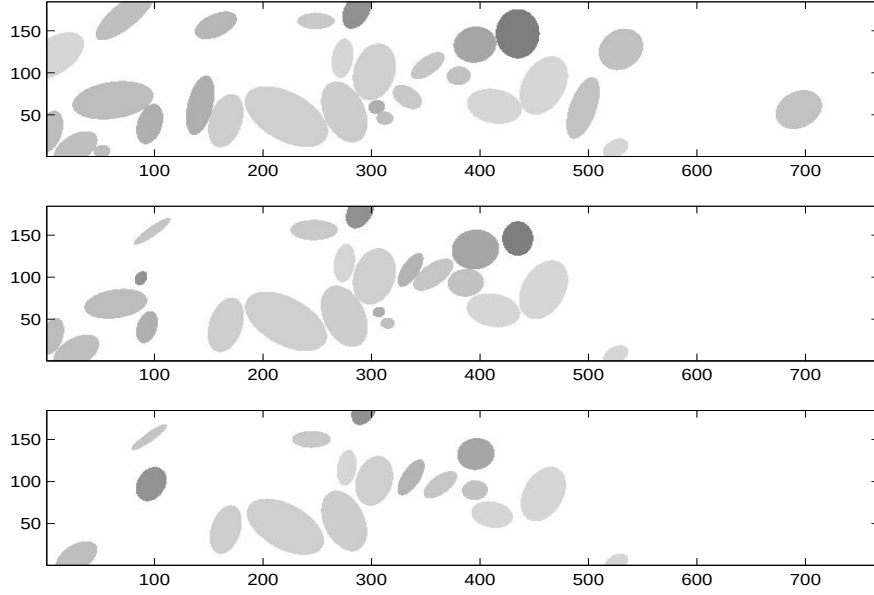


FIGURE 4: 2D sections orthogonal to the z -axis for $z = 3$ (top), $z = 7$ (middle) and $z = 12$ (bottom) after 50000 MCMC iterations. The grey level of each cell represents the fluorescence level λ .

images with a different pattern of cell configurations. Whatever choice is made, the acceptance probabilities $\alpha_\nu(x, x')$ and $\alpha_{\nu'}(x', x)$ automatically account for the proposal distribution used and ensure the MCMC sampler has the correct limiting distribution.

4.2 INITIAL RESULTS

Applying the MCMC algorithm as defined above does not yield very satisfactory results. This is due to the difficulty in reaching a cell configuration that fits the data at all well. The problem of convergence for the algorithm is well illustrated in Figure 4, which shows the configuration reached in 50000 iterations after starting from a blank scene. Comparisons with the original data show the sampler has managed to locate many of the cells in the correct places, but some are missing and there are cases where a single large cell appears where there should be several small cells. Although some improvements are possible by fine-tuning elements of the algorithm, the basic problem of failure to find a completely plausible configuration remains.

Given the heavy computation involved in reaching these imperfect configurations, there is good reason to consider other ways of initialising the MCMC algorithm. If a better initial configuration can be found by other methods, the MCMC algorithm can be started at this state and, after a relatively short burn-in period, it will generate realisations that can be used to draw inferences.

4.3 AN MCMC ALGORITHM USING A GOOD STARTING STATE

We first describe a fully automated way of finding a good initial state for the MCMC algorithm. The procedure comprises two steps.

- Step 1: Generate an initial segmentation using morphological operators, as described in Section 2.2. Figure 3 illustrated the segmentation achieved by doing this.
- Step 2: For each set of connected voxels, fit an ellipsoid and thereby define a cell. In fitting the ellipsoid, we treat co-ordinate vectors of individual voxels as a set of observations from a 3D normal density, define the centre of the ellipsoid as the sample mean, and obtain values for the semi-axes and angles of rotation by matching the sample variances and covariances. The cell's fluorescence intensity is computed from the average record inside the ellipsoid. The result of applying this step in our example is shown in Figure 5.

From this starting point, we run the MCMC algorithm, as originally defined, to give a more accurate fit to the observed data. This may involve the merging of some cells or creation of others. Running 50000 iterations from the morphology-based starting configuration produced the image shown in Figure 6, where it can be seen that new cells have indeed been created and cell positions have changed to give a better fit to the data. Comparison with Figure 1 shows a good match to the observed data and, after discarding the first part of the run as burn-in, these samples could be used for inference.

Since the MCMC sampler retains all of the move types previously specified, cells can still appear and disappear over time and it is desirable that this should happen if there is uncertainty in the posterior distribution about the total cell count. In our

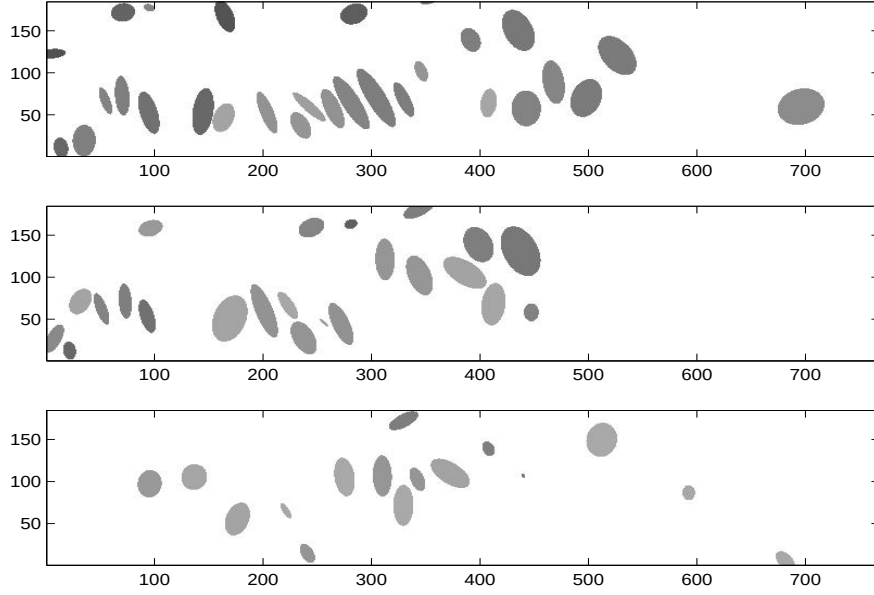


FIGURE 5: Optical sections at $z = 3, 7$ and 12 showing 3D configurations after converting sets of connected voxels from the morphological analysis into ellipsoids.

particular example, the observations provide quite clear information about the cells present and, once identified, these cells they can be expected to remain in place.

4.4 USER INTERVENTION TO CONSTRUCT A GOOD STARTING STATE

In the past, data of the type considered here have often been analysed with substantial input from “expert users”. Such expertise can be very helpful but it is desirable to keep to a minimum the degree of assistance required from such an expert. Our suggestion in this regard is to construct a morphology-based starting point as in Section 4.3, then run a moderate number of iterations of MCMC. Having defined a reasonably good configuration, we then ask the expert user to indicate any corrections deemed necessary. This can be done by identifying a point inside each cell which the expert believes should be present and indicating the need for new cells by specifying a point in each of these. The points specified for each cell do not need to be at the cell centres but, particularly for the additional analysis of Section 4.5, it is helpful if the point is safely in the cell interior. Rejected cells are deleted. New cells can be allocated randomly generated properties, which are then updated by a short series of MCMC steps applied

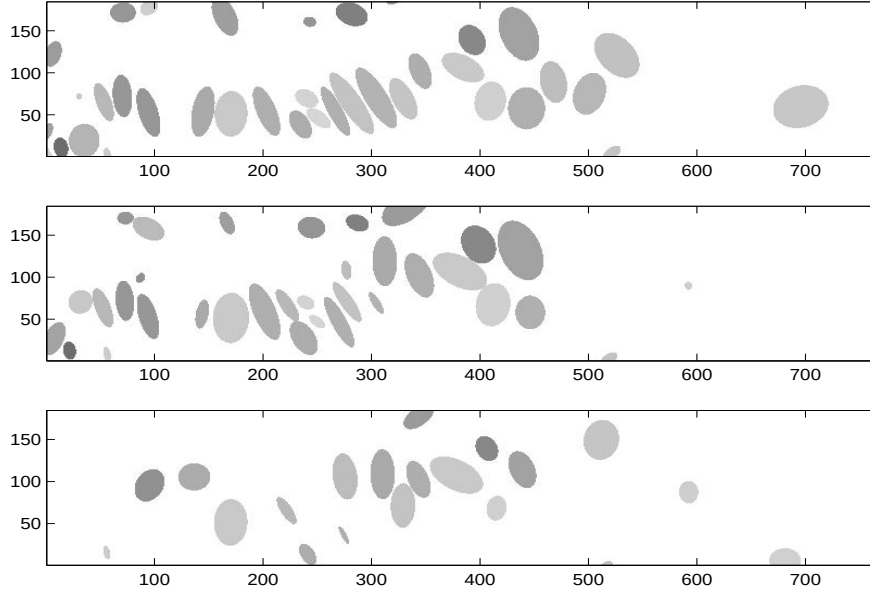


FIGURE 6: A sample after 50000 iterations of the MCMC sampler using the initial state derived from the morphological analysis.

to these parameters alone, to fit them to the data. This provides the starting point for the standard MCMC algorithm and, after a suitable burn-in period, samples can be used for inference.

The MCMC algorithm output given in Figure 6 was used as input for user intervention. The configuration after addition and deletion of cells by the user, but before any further updating, is seen in Figure 7.

4.5 A CONDITIONAL ANALYSIS

In general, it is not possible when using MCMC methods to study attributes of a particular cell as cells do not have any form of permanent identity. Updates of location may cause a cell to move to a different part of the scene or it may disappear altogether in a merge or death move.

Suppose, however, that after initial processing and user intervention, one is confident that all cells are correctly identified (this may not be possible if the degree of blurring or signal noise is too high). MCMC sampling may be continued under the condition that the number of cells remains fixed and each cell is constrained to

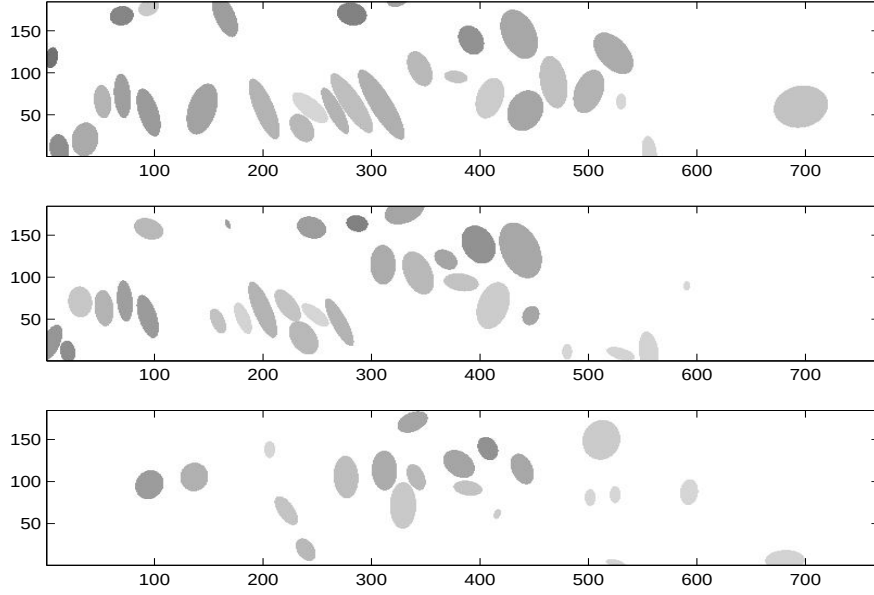


FIGURE 7: The object configuration after finding an initial state by morphological analysis, running the MCMC sampler for 50000 iterations, and allowing user intervention

retain the user-identified point in its interior. The first condition is met by dropping birth, death, split and merge moves and the second condition can be implemented by rejecting proposals that would violate it. Effectively, these two constraints modify the posterior distribution and the samples obtained are from the conditional version of this distribution, given the imposed conditions. If signal noise is low and we are correct in assuming cells have been correctly identified, there should be little difference between the conditional and unconditional posterior distributions. A sample after running this constrained MCMC algorithm is shown in Figure 8.

5 IMAGE INTERPRETATION

5.1 ESTIMATING PROPERTIES OF THE CELLS BY MCMC

Cartilage tissue can be categorised as belonging to four different zones (Woessner and Howell, 1993), the tangential, middle and deep layers, and the calcified cartilage layer. The middle and lower layers occupy 80% to 90% of the tissue and can be distinguished

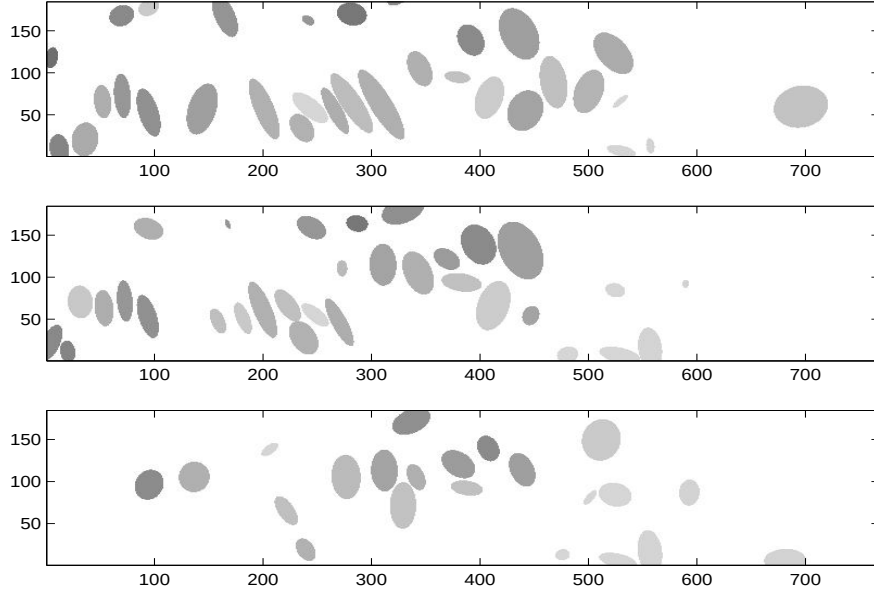


FIGURE 8: 2D optical sections at depths $z = 3, 7$ and 12 after running the conditional MCMC algorithm from the user-refined starting point.

by the types of cells they contain. The middle layer contains more rounded cells known as chondroblasts, while the deep layer contains more elliptically shaped cells, chondrocytes, arranged in columns. Properties such as the relative frequencies of the two types of cell in a region and their shape, size and fluorescence intensities are informative criteria. These aid in understanding the processes in the cartilage tissue and provide information on the proliferation and differentiation of various cellular components (Noda, 1993).

In order to distinguish cell types in the marked point process model, we add the label of type I for a chondroblast cell or type II for a chondrocyte as a further element of the cell's mark. With a mixture model for the mark distribution, cells can be attributed properties conditional on their type. In the reference model underlying our prior distribution, we assume cells are equally likely to be of types I and II. Given the cell type, the semi-axes are assumed to be independent and follow normal distributions constrained to positive values. The information that type II cells are more spherical than type I cells is incorporated in the means of these three distributions, set as 30, 12.5 and 4 for type I cells and 20.5, 17.5 and 4 type II cells. In both cases, the variance

is 4.5^2 . The distributions for orientation and fluorescence of both cell types remain as before. These features of the reference model carry through to the prior itself with minor perturbations following from the constraint in (2) that cells cannot overlap.

The Markov chain Monte Carlo algorithm is extended to accommodate cell type by adding a relabelling move which can change a cell's label while holding other values fixed. With this additional move, the conditional sampler was applied and run for 100000 iterations. We started from the configuration in Figure 8 with arbitrarily assigned cell types. After deleting the first 10% of the iterations to allow for burn-in, measurements were made on the samples. Information was extracted concerning the size and shape attributes of the two cell types. Point estimates and credible regions for quantities of interest can be found by taking ergodic averages over the observed samples. The variance of the sample averages is inflated by the integrated autocorrelation time, which we denote by τ , so the standard error from a sample of size N for an attribute with variance σ^2 is $\sqrt{\tau\sigma^2/N}$. The quantities σ^2 and τ can be estimated from the samples: see Green and Han (1992) and Geyer (1991). Figure 9 shows a sample after 100000 iterations with the current labelling of each cell indicated. We discarded the first 10000 iterations, then subsampled the Markov chain once every 100 iterations. Figure 10 shows histograms of various attributes for the two types of cell. The estimated proportion of type I cells is 0.442. The estimated attribute means, together with standard errors and integrated autocorrelation time τ for the subsequence are given in Table 1.

6 CONCLUSIONS

We have shown that three-dimensional confocal microscope data of cartilage cells can be analysed using a marked point process model. Additional features such as a variety of cell types are readily included in such a model.

With each cell covering a sizeable number of voxels, a single iteration of the MCMC algorithm requires significant computation. Furthermore, progress within the state space tends to be quite slow, particularly for birth, death, merge and split moves where the step, or its converse, relies on a randomly generated, high dimensional

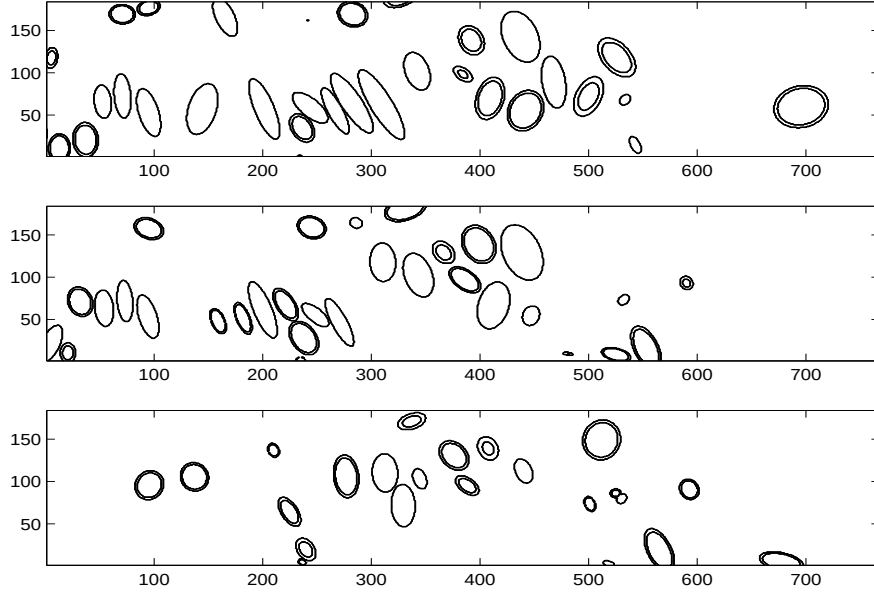


FIGURE 9: A sample realisation labelling the cells as type I (single-line border) and type II (double-line border).

proposal fitting the recorded data well. Our morphology-based scheme for producing a good initial state avoids the excessively long burn-in period that would otherwise be necessary. Adding a small amount of input from the user improves this scheme further. This allows computational effort to be targeted on the task of sampling the posterior distribution in the part of the sample space where it is concentrated.

Our general strategy would appear to have wider applicability to many sampling problems where the target distribution is focused on a small fraction of a high-dimensional sample space and this region is not easily found by the standard MCMC algorithm. Care is needed, however, if the target distribution is multi-modal since a long sojourn in a single mode may give the impression the Markov chain is mixing much more effectively than it really is. In our example, modes of the posterior distribution correspond to different subsets of cells being present in the configuration and the user can check that samples are in agreement with the observed data. In other applications, further diagnostic checks may well be advisable.

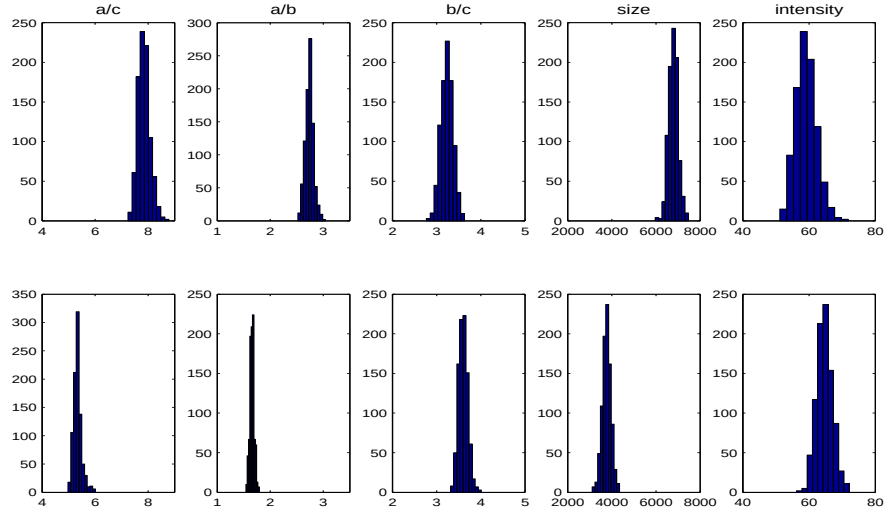


FIGURE 10: Histogram of convexity, volume and intensity for type I cells (top) and type II cells (bottom).

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Attribute	Type	Mean	S.E.	τ
a/c	I	7.8	0.5	4.4
	II	5.3	0.7	19.7
a/b	I	2.7	0.3	14.6
	II	1.7	0.14	12.0
b/c	I	3.2	0.3	4.8
	II	3.6	0.4	12.4
Volume	I	6790	530	5.8
	II	3770	290	8.8
Intensity	I	59	11	13.2
	II	65	16	44.0

TABLE 1: Estimated attributes for the two types of cell with standard errors and integrated autocorrelation times.

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